Owner's Manual

F Series Dock Leveller

Model No: DL714

Static Capacity: 12,500kg

Version no: 1

September 2016

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Standard Specifications

Safetech Hydraulic Dock Levellers					
Model	DL703-F	DL708-F	DL710-F	DL712-F	DL714-F
Deck Width	2110mm	2110mm	2110mm	2110mm	2110mm
Deck Length	600mm	2210mm	2820mm	3440mm	4040mm
Lip Length	490mm	490mm	490mm	490mm	490mm
Nominal Overall Travel Up	350mm	750mm	750mm	1000mm	1075mm
Nominal Overall Travel Down	75mm	180mm	180mm	180mm	180mm
Nominal Travel Up At 10% Gradient	60mm	210mm	275mm	335mm	400mm
Nominal Travel Up At 15% Gradient	N/A	310mm	410mm	500mm	600mm
Static Capacity	7000kg	12,500.00kg			
Type of Installation	Pit Mounted	Pit Mounted or Tailgate Recess			
Pit Types available	Pit Mounted- Available on all models Tailgate Recess- Available on 708, 710, 712 & 714 Step Down - Available on 710 & 714 Only Ezy Pit – Standard -Available on 710 & 714 Only Ezy Pit – Tailgate - Available on 710 & 714 Only				
Dock Bumpers	Rubber (Standard), Composite (Optional)				
Pit Surround	Galvanised (Supplied by Safetech, installed by others)				
Finish	Dock leveller is shot blasted, primed with rust inhibiting primer and finished with two (2) coats of industrial enamel. Colour – Blue RAL 5010. Dry Film Thickness – 40 microns				
Toe Guards	Telescopic guards on both sides, painted safety yellow				
Power pack	1.1kW 415V 3 Phase 50Hz Mounted to base frame underneath the deck				
Controls	Low voltage 24V wall mounted push button				
Power Requirements	415V (3 Phase) 10amp 50Hz outlet in pit and Isolating switch on wall by others				
Operation	Constant Pressure Single Push Button Control for operation of Deck Platform and Loading Lip. Incorporating Full Hydraulic System with Hydraulic Fail Safe Valve on Deck Cylinder.				
	Dock Levellers is fitted with the Intrinsically Safe Safety Prop				
	The Safety Prop will support both the Lip and Deck and can withstand A 4,500 Kg Moving Load and has a Lockout / Tag-Out Capability.				
	The Safety Prop is activated from above the platform and is a one man operation to both engage and disengage.				

1.0 Introduction

1.1 Scope of this Manual

This manual covers the operation of the Dock Leveller after it has been installed and commissioned.

Do not attempt to install or commission a Dock Leveller unless you have been trained and authorised by Safetech.

1.2 Disclaimer

Although every effort is made to ensure accuracy, currency and completeness of the information contained in this manual, Safetech does not guarantee, warrant, represent or undertake that the information is correct, accurate or current.

Safetech is not liable for any loss, claim, cost or expense of any kind arising in any way whatsoever from the information contained in this manual.

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1.4 Owner's Responsibility

To ensure both your safety and the safety of others, and the continued effective operation of your Safetech Dock Leveller, your responsibilities are as follows:

- Before operating the Dock Leveller, please read and ensure you understand both the safety instructions and the operating instructions.
- If you do not fully understand these instructions, contact either Safetech or an authorised Safetech agent for clarification before operating your Dock Leveller.
- The Owner must review the Risk Assessment in this manual.
- Please Note: That in the Risk Assessment a number of items have been flagged as Completed? = No. The Owner must include these hazards in their own operational risk assessment and ensure that action is taken to address the hazard.
- The owner is responsible to ensure all operators have been adequately trained on the safe operation of the Dock Leveller.
- The owner is responsible to ensure that a risk assessment on the Dock Leveller in their operating environment is done.

2.0 Safety Instructions

Before operating the dock leveller, please read and ensure you understand both the safety instructions and the operating instructions.

If you do not fully understand these instructions, contact either Safetech or an authorised Safetech agent for clarification before operating your dock leveller.

- Improper operation of this dock leveller may result in serious personal injury. Do not operate unless you have been properly trained and have read, and understand both the safety and operating instructions.
- DO NOT allow any unauthorised access to, or under the dock leveller.
- Ensure all personnel are clear of the boom gate when the boom gate is operating.
- DO NOT allow any person near the dock leveller when it is operating or near a truck when it is reversing toward, or leaving the dock.
- DO NOT exceed the rated capacity as indicated on the serial-plate.
- DO NOT allow any equipment on the Dock Leveller when the truck is leaving the dock.
- DO NOT attempt to lift the Dock Leveller with personnel or equipment on the unit.
- DO NOT drive loading equipment (eg forklift) onto the Dock Leveller when the Dock Leveller is in the below cross traffic position and is NOT resting on the truck. Ensure load lip rests securely on the truck bed during loading / unloading operations.
- Trucks should be restrained by an effective restraint system during loading / unloading operations to prevent movement of the truck.
- DO NOT use a damaged or broken Dock Leveller.
- DO NOT drive loading equipment close to the edge of the Dock Leveller.
- DO NOT operate Dock Leveller if the height of the truck bed is more than 14% above the cross traffic position or more than 3% below the cross traffic position.
- Excessive incline of the Dock Leveller may result in unsafe operating condition for a loading equipment.
- Loading equipment should not exceed 5 kph whilst on the Dock Leveller.
- DO NOT operate Dock Leveller if under the influence of drugs or alcohol.
- If you have an ‘AUTO RETURN TO CROSS TRAFFIC’ model, when the truck departs the dock will automatically return to the ‘cross traffic’ position. You need to :
- Ensure no personnel are in the area of the truck and the dock leveller when the truck departs.
- Do a risk assessment and develop and implement a safe working procedure
- Train your personnel on the safe work procedure
- When not in use the Dock Leveller should be in the cross traffic position with the lip tucked inside the base frame front angle.

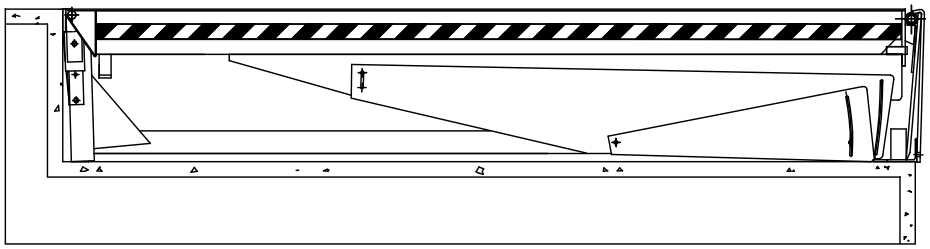


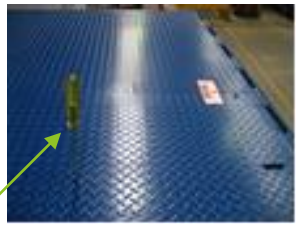
Figure 1: Dock Leveller in cross traffic position with lip tucked in

- DO NOT enter under the dock leveller for cleaning unless you have been trained and authorised to do so.
- The owner must also do a risk assessment and develop safe work procedures for ergonomically sound cleaning practices.
- For routine cleaning, the Dock Leveller must be securely resting on the maintenance prop as per the following instructions, the area must be cordoned off with a suitable safety barrier to prevent access by personnel or equipment and the main power to the Dock Leveller must be isolated by an effective Lock out / Tag out system.
- Clean from outside the dock using a long handled cleaning device.
- •DO NOT spray electrical system with high pressure water.

Dock levellers fitted with the Intrinsically Safe Maintenance Prop

2.1 To Engage the Maintenance Prop:

Engaging the maintenance prop is a one man operation.



- With the dock leveller in the parked position, stand on top of the dock leveller.
- Insert a screwdriver (approx 250mm long) or similar device through the hole in the deck and push down on the maintenance prop lever. The screwdriver needs to be inserted approximately 40mm. The tension spring will then pull the safety lever down and the maintenance prop will be activated.



- Stand clear of the dock leveller and raise the deck until the lip fully extends. When the deck is fully raised and the lip fully extended, release the dock leveller raise button. The lock out section of the maintenance prop will automatically protrude through the hole in the deck as the deck lowers onto the maintenance prop.



- The maintenance prop should now be automatically engaged with the lip fully supported.



- Lock out the dock leveller by putting a pad lock though the hole in the tongue that protrudes through the deck. Take care when walking on the raised deck.
- Isolate the main power to the dock using an effective Lock out/Tag out system.



2.2 To Safety Release the Maintenance Prop:

- Stand at the front of the dock leveller at ground level. Firmly push back on the release arm until the release arm hits the underside of the deck. Note: When pushing back the release arm by hand take care as the release arm is under spring tension and if not firmly pushed, the arm may pull back under tension and the hand may hit on the maintenance prop. A force of approx 20 kg is required to push the release arm back.



- After activating the release arm stand on top of the deck and remove the padlock from the tongue protruding through the deck. Take care when walking on raised deck.
- Remove Lock out /Tag out system.



- From the control station fully raise the deck then release the raise button. The maintenance prop will automatically disengage as the deck begins to lower. Lower the deck to the parked position
- The dock leveller is now ready for normal use.
- Maintenance and service on all dock levellers must only be carried out by a person who has been trained and accredited by Safetech, or has been trained and authorised by the Owner. A Safetech Service maintenance prop, or equivalent, properly engaged must be used when doing service or maintenance.



Safetech Service Maintenance Prop

- DO NOT enter under the Dock leveller for service or maintenance purposes unless you have been trained and authorised to do service work. A Safetech Service maintenance prop, or equivalent, properly engaged must be used when doing service or maintenance work.
- Maintenance and service on the dock leveller must be done at regular intervals to ensure trouble free and safe operation. For high cycle or heavy load environments, service should be done every 3 months. For medium or low cycle environments, service should be done every six months.
- If the dock leveller is not operating correctly or is undergoing maintenance or service work, an effective 'LOCKOUT / TAG OUT' procedure must be used.
  eg: Lock-out isolating switch with an assigned individual key-padlock and danger tag to prevent the possibility of inadvertent energisation.

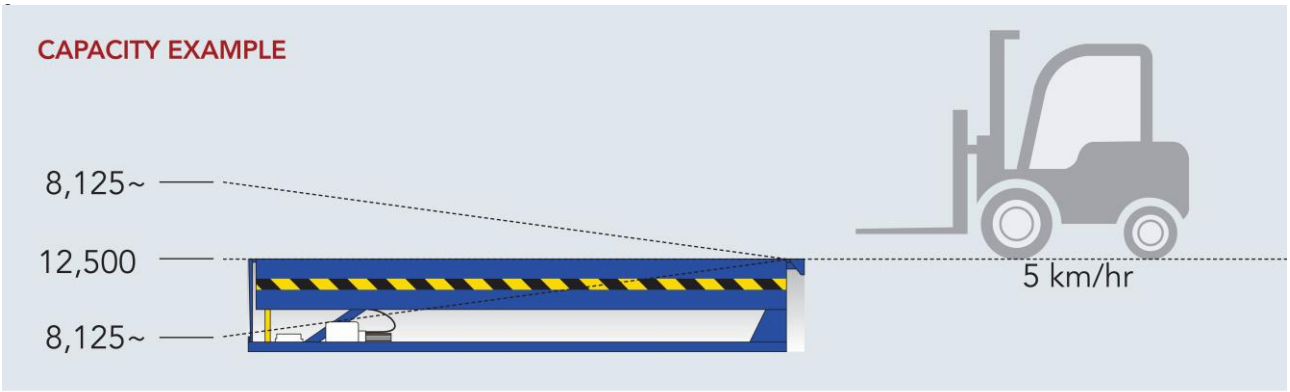
Key Padlock and Danger Tag MUST be used



- DO NOT modify, dismantle or dispose of the Dock Leveller unless you have been trained and authorised to do so.

2.3 Static and Dynamic Load Capacities

- To assess the required capacity, all Safetech Dock Levellers have a rated static capacity, which is measured when the Dock Leveller is stowed.
- Safetech Dock Levellers are market leaders in their safe operating dynamic load capacities, which is approximately 65% of the rated static capacity.
- These measurements assume that any forklift or loader does not exceed 5km/hr and the operating grade does not exceed 10%.
- For conditions which exceed these standards, please contact Safetech for your individual requirements.

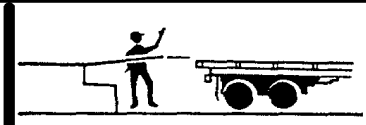
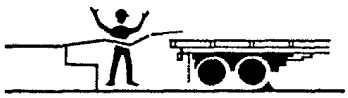
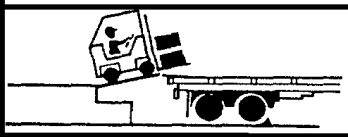
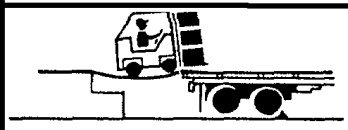
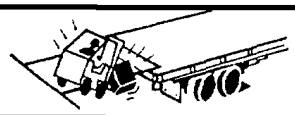
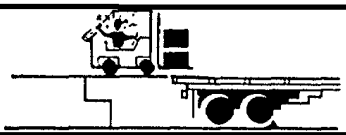


** Higher capacity levellers available based on gross fork / load weights, servicing gradients, speed and usage frequency*

2.4 Safety Instructions

Abide by the requirements defined in the 'Good Safety Practices' sign when operating the dock leveller

Good Safety Practices

STAY CLEAR OF DOCK LEVELLING DEVICE WHEN FREIGHT CARRIER IS ENTERING OR LEAVING AREA.	
CHOCK/RETRAIN ALL FREIGHT CARRIERS.	
DO NOT USE BROKEN OR DAMAGED DOCK LEVELLING DEVICE.	
DO NOT MOVE OR USE THE DOCK LEVELLING DEVICE IF ANYONE IS UNDER OR IN FRONT OF IT.	
KEEP HANDS AND FEET CLEAR OF PINCH POINTS.	
LIP MUST REST SECURELY ON FREIGHT CARRIER BED.	
DO NOT USE DOCK LEVELLING DEVICE IF FREIGHT CARRIER IS TOO HIGH OR TOO LOW.	
DO NOT OVERLOAD THE DOCK LEVELLING DEVICE.	
KEEP A SAFE DISTANCE FROM BOTH SIDE EDGES.	
DO NOT OPERATE ANY EQUIPMENT WHILE UNDER THE INFLUENCE OF ALCHOL OR DRUGS.	
DO NOT LEAVE EQUIPMENT OR MATERIAL UNATTENDED ON DOCK LEVELLING DEVICE.	

3.0 Operating Instructions

Before operating the Dock Leveller

Please read and & ensure you understand

Both the Safety Instructions & the

Operation Instructions applicable to your unit.

If you do not fully understand these

Instructions, contact either Safetech or an

Authorised Safetech Agent for clarifications

Before operation your Dock Leveller

Caution

When access to the Underside of the

Dock Leveller

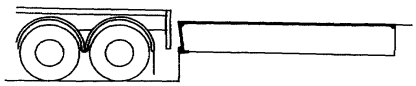
Is required for cleaning or any other

Purpose refer to the Safety Instructions

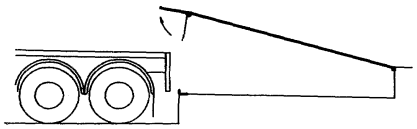
3.1 Operating Instructions with Single Push Button

WARNING: Do not exceed 14% Gradient on Deck

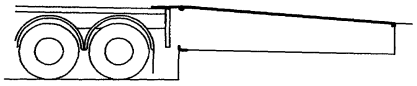
IMPORTANT: - STAND CLEAR WHEN OPERATING UNIT



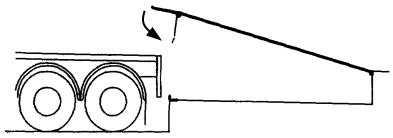
1. Open vehicle loading hinge doors, reverse vehicle into position against dock bumpers.



2. Press dock leveller **RAISE** button until deck is raised and lip is fully extended.



3. Release **RAISE** button. Dock leveller will lower until the loading lip rests on vehicle floor. When unloading, as vehicle floor raises with decreased load, deck will automatically adjust upwards to loading height. When loading, as vehicle floor lowers with increased load, deck will readjust to new lowered height.



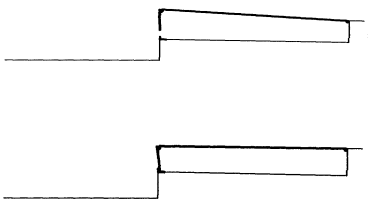
4. PRIOR to vehicle leaving the dock, press **RAISE** button. As the deck raises the lip will lower into the vertical position. HOLD button only until the lip is completely vertical and clear of any obstruction



5. Release the **RAISE** button to allow deck to float into the cross traffic position.



6. If the dock leveller is below level, to shut away press the **RAISE** button until deck lifts to approximately 300mm (12”) above horizontal.



7. Release **RAISE** button and dock leveller will return to cross traffic position.

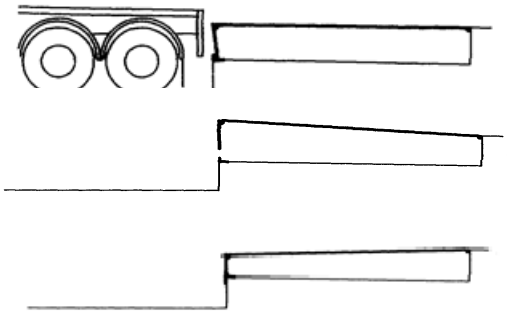
NOTE:

- If the dock leveller has been servicing vehicle below dock height, follow instructions 6 & 7 to return it to cross traffic position.

3.2 Operating Instructions with Independent Lip Control

WARNING: Do not exceed 14% Gradient on Deck

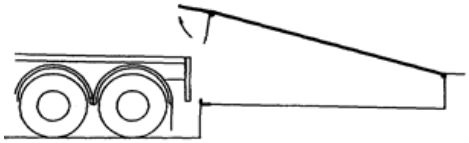
IMPORTANT: - STAND CLEAR WHEN OPERATING UNIT



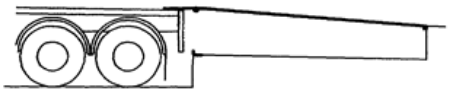
1. Reverse vehicle into position against dock bumpers.

2. Press and hold the dock leveller "dock raise" button. While the dock leveller is raising, jog the lip out to clear the lip supports using the independent lip control. Release the dock leveller "dock raise" button and the dock leveller will lower to the lowered position.

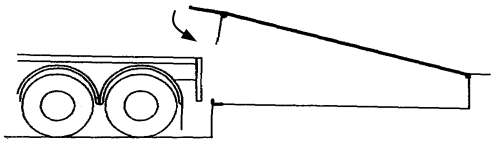
3. Open vehicle hinge doors.



4. Press dock leveller raise button until lip clears rear of vehicle. Whilst holding down the “dock raise” button, simultaneously press the "lip raise" button until the lip fully extends.



5. Release “dock raise” button. Dock leveller will lower until the
Loading lip rests on vehicle floor. When unloading, as vehicle Floor raises with decreased load, deck will automatically adjust upwards to loading height. When loading, as vehicle floor lowers with increased load, deck will readjust to new lowered height.



6. Prior to vehicle leaving the dock, press “dock raise” button. As the deck raises the lip will lower into the vertical position. HOLD button only until the lip is completely vertical and clear of any obstruction.



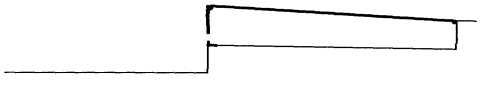
7. Release the “dock raise” button to allow deck to fall to cross traffic position.



8. To close vehicle hinge doors, press and hold the dock leveller "dock raise" button. While the dock leveller is raising, jog the lip out to clear the lip supports using the independent lip control. Release the dock leveller "dock raise" button and the dock leveller will lower to the lowered position.



9. Close the vehicle hinge doors.



10. To shut away press the “dock raise” button until deck raises to approximately 300mm (12”) above horizontal.



11. Release “dock raise” button and dock leveller will return to cross traffic position.

4.0 Maintenance

4.1 Maintenance & Repair Personnel:

Safetech Equipment should be serviced and maintained by a qualified a Safetech representative. If you require servicing you can entering into a service agreement with Safetech. Please click on the link to obtain the Safetech Service Agreement. http://www.safetech.com.au/safetech-service-agreement

If this is not possible, then the Owner must:

- Develop & implement a preventive maintenance plan for the unit.
- Create a competency based training program for servicing the unit.
- Train & authorise personnel on the safe & correct service of the unit.
- Permit only suitably trained & skilled personnel to service the unit.

4.2 Planned Maintenance:

To prolong the operational life and ensure the safe operation of you Dock Leveller table we recommend regular planned maintenance by Safetech or a Safetech authorised agent.

For high cycle or heavy load environments, service should be done every three (3) months. For medium or low cycle environments, service should be done every six (6) months.

Please contact either Safetech or a Safetech authorised agent to discuss your maintenance requirements.

4.3 Troubleshooting:

If, for any reason your Dock Leveller does not work, check the following:

- Is the power supply switched on?
- Has the Dock Leveller been operated with loads in excess of the stated capacity?

If the power supply is switched on and the Dock Leveller has not been operated with loads in excess of the stated capacity, and the Dock Leveller does not work, contact or an authorised agent to rectify the problem.

Damage to the Dock Leveller or lack of prescribed maintenance may result in the Dock Leveller being inoperable.

For Repairs or Services or a Planned Maintenance Program:

Contact Safetech on 03 8546 5600

5.0 Warranty

Refer to Safetech complete Terms of Sale and Warranty at http://www.safetech.com.au/terms-sale-warranty/

6.0 Safetech' Divisions

Head Office	
Head Office – Moe, Victoria	**Address:** 39-45 Della Torre Road, Moe, Victoria, 3825 **Postal Address:** PO Box 360, Moe, Victoria, 3825 **Phone:** 1800 674 566 Email: sales@safetech.com.au
Service	
Service Enquiries	**Phone:** 03 8546 5600 Email: service@safetech.com.au
Sales offices	
Victoria	**Address:** 1a/21 Lionel Road Mt Waverley VIC 3149
New South Wales	**Address:** 14a/65 Elizabeth Street Wetherill Park NSW 2164
Queensland	**Address:** 1/23 Armada Street, Banyo, QLD, 4015
Western Australia	**Address:** 11/288 Amherst Road Canning Vale, WA 6155

7.0 Risk Management

7.1 Risk Assessment:

Safetech Hydraulic Dock Leveller						
Hazard No	**Hazard / Risk**	Probability	Consequence	**Control**	**Risk control implementation**	
					Responsibility of:	Completed ?
1	Inadequate information supplied to enable the safe installation of the Dock leveller.	Possible	Catastrophic	• Provide Installation Manuals to Installers that have all the necessary information to safely install the unit and ensure that it operates correctly.	Product designer	Yes
2	Untrained personnel installing the equipment could result in injury to personnel and damage to equipment.	Possible	Catastrophic	• Read & understand the Installation manual. • Create a competency based training program for installation personnel. • Train & authorise installers on the safe & correct installation of the unit using a Safetech Service maintenance prop or equivalent. • Only suitably trained & skilled personnel to install unit.	Installer	Yes if Safetech. No if not Safetech.
3	Inadequate information supplied to enable the safe operation of the Dock Leveller could result in damage to equipment and injury to personnel.	Possible	Catastrophic	• Provide Owners Manuals, Operating Instructions & Signage to the Owner that have all the necessary information for people to safely operate, maintain in good condition during it's operating life and then finally dispose of the equipment after it's useful life.	Product designer	Yes
4	Untrained operators using the Dock Leveller could result in damage to equipment and injury to personnel.	Possible	Catastrophic	• Read & understand the Owner's Manual. • Create an adequate competency based training program for operators • Train & authorise operators on the safe & correct use of the unit. • Only trained & authorised personnel to use Dock leveller.	Owner	No

5	Untrained personnel attempting to service or do maintenance on a Dock Leveller could result in damage to equipment or injury to personnel. Also refer to Hazard Nos 15, 16 & 17 (Hydraulic & Electrical)	Possible	Catastrophic	• Owner to ensure only Safetech trained and accredited people carry out service or maintenance. OR • Develop & implement a preventive maintenance plan for the unit. • Create a competency based training program for servicing the unit. • Train & authorise personnel on the safe & correct service of the unit using a Safetech Service maintenance prop or equivalent. • Only suitably trained & skilled personnel to service the unit.	Owner	No
6	Untrained personnel modifying the Dock Leveller could result in injury to personnel and damage to equipment.	Possible	Catastrophic	• Owner to ensure only Safetech trained and accredited people modify the unit. OR • Develop & implement a plan for modifying the unit. • Create a competency based training program for modifying the unit. • Train & authorise personnel on the safe & correct modification of the unit using a Safetech Service maintenance prop or equivalent. • Only suitably trained & skilled personnel to modify the unit.	Owner	No
7	Untrained people dismantling the Dock Leveller could result in damage to equipment and injury to personnel.	Possible	Catastrophic	• Owner to ensure only Safetech trained and accredited people dismantle the unit. OR • Develop & implement a plan for dismantling of the unit. • Create a competency based training program for personnel dismantling of the unit. • Train & authorise personnel on the safe & correct dismantling of the unit using a Safetech Service maintenance prop or equivalent. • Only suitably trained & skilled personnel to dismantle the unit.	Owner	No
8	Untrained people disposing of the Dock Leveller could result in damage to equipment, injury to personnel and non-compliance to regulations.	Possible	Catastrophic	• Owner to ensure only Safetech trained and accredited people dispose of the unit. OR • Develop & implement a plan for disposal of the unit. • Create a competency based training program for personnel disposing of the unit. • Train & authorise personnel on the safe & correct disposal of the unit using a Safetech Service maintenance prop or equivalent. • Only suitably trained & skilled personnel to dispose of the unit.	Owner	No

9	Inadequate service & maintenance of the unit results in unsafe operation and potential injury to personnel.	Possible	Catastrophic	• Contact Safetech for the regular servicing of your Dock leveller. (3 monthly service intervals for a heavy workload unit & 6-month service intervals for a low workload unit.) OR • Develop & implement a preventive maintenance plan for the unit. • Create a competency based training program for servicing the unit. • Train & authorise personnel on the safe & correct service of the unit using a Safetech Service maintenance prop or equivalent. • Only suitably trained & skilled personnel to service the unit.	Owner	No
10	Unknown safety hazards are existing in the area due to unknown local practices or the impact of the Dock Leveller in the area could result in injury to personnel and damage to equipment.	Possible	Catastrophic	• Owner must ensure that a risk assessment on the unit in their operating environment is done	Owner	No
11	When the Dock Leveller is above the cross traffic position there is a risk of a shear/crush injury between the Dock Leveller sides & the dock pit edge.	Possible	Major	• Side Safety Guarding fitted to sides of Dock Leveller to prevent foot/hand intrusion. • Provide clear instructions in the Owner's manual that specify that the unit must only be operated when all personnel are clear of the Dock Leveller area.	Designer	Yes
				• Ensure that the unit is only operated when all personnel are clear of the dock area. • Include this hazard in your risk assessment.	Owner	No
12	If side safety guards are missing or damaged there is a risk of a shear/crush injury between the Dock Leveller sides & the dock pit edge.	Possible	Major	• Provide clear instructions in the Owner's manual that specify that the unit must only be operated when all personnel are clear of the Dock Leveller area.	Designer	Yes
				• Ensure that the unit is only operated when all personnel are clear of the dock area. • Include this hazard in your risk assessment.	Owner	No
13	When the Dock Leveller is lowered onto the truck bed there is a risk of a shear/ crush injury at contact point between load lip and vehicle floor.	Possible	Major	• Provide clear instructions in the Owner's manual that specify that the unit must only be operated when all personnel are clear of the Dock Leveller area.	Designer	Yes
				• Ensure that the unit is only operated when all personnel are clear of the dock & truck. • Include this hazard in your risk assessment.	Owner	No

14	With one button operation, when the raise button is released the Dock Leveller automatically lowers to a resting point (either truck bed or lowered position), and there is a risk of a shear /crush injury at any contact point.	Unlikely	Major	• Provide clear instructions in the Owner's manual that specify that the unit must only be operated when all personnel are clear of the Dock Leveller area.	Designer	Yes
				• Ensure that the unit is only operated when all personnel are clear of the dock area. • Include this hazard in your risk assessment.	Owner	No
15	When Dock Leveller is being moved to the cross traffic position or fully lowered position there is a risk of a shear/crush injury at pivot edge of Dock Leveller and load lip.	Unlikely	Major	• Provide clear instructions in the Owner's manual that specify that the unit must only be operated when all personnel are clear of the Dock Leveller area.	Designer	Yes
				• Ensure that the unit is only operated when all personnel are clear of the dock area. • Include this hazard in your risk assessment.	Owner	No
16	Electrical Hazard **(**: 240V and 3 phase 415V**)** created if power cable damaged could result in shock, burns or seizure to personnel.	Rare	Catastrophic	• Wiring to relevant Aust. Stds. & to the requirements of the Chief Electrical Inspector	Designer	Yes
17	Operators in contact with 3 phase 415V supply voltage could suffer shock, burns or seizure.	Rare	Catastrophic	• 3 Phase, 415V supply outlet mounted to rear of pit to be inaccessible to un-authorised persons. • Supply to outlet run in protective conduit.	Designer	Yes
				• Wiring complies with all Relevant Standards and requirements.	Installer	Yes if Safetech. No if not Safetech.
18	Hydraulic failure allowing collapse of unit can result in injury to personnel under or near Dock Leveller.	Rare	Major	• Fit a Velocity Control device to prevent sudden collapse of unit, and ensure specifications of manufacture comply with relevant standards. • Provide clear instructions in the Owner's manual that specify that the unit must only be operated when all personnel are clear of the Dock Leveller area. • Provide clear instructions in the Owner's manual that specify the requirements for access under the unit.	Designer	Yes
				• Develop & implement a safe work procedure for cleaning under the dock. • Include this hazard in your risk assessment.	Owner	No
19	Load Lip collapse due to excessive load placed on the lip could result in injury to personnel and damage to equipment.	Rare	Catastrophic	• Provide clear instructions in the Owner's manual on load limits.	Designer.	Yes
				• Do not overload the Dock Leveller. • Include this hazard in your risk assessment.	Owner.	No

20	Structural damage to Dock Leveller if dock subjected to loads greater than the specified capacity could result in damage to equipment and injury to personnel.	Rare	Catastrophic	• Provide clear instructions in the Owner's manual on load limits.	Designer	Yes
				• Do not overload the Dock Leveller. • Include this hazard in your risk assessment.	Owner	No
21	Damage to Power Pack and/or Control Box could result in injuries to personnel from both exposure to live power and equipment failure.	Rare	Catastrophic	• Install Power pack and control box as specified by Designer.	Installer	Yes if Safetech. No if not Safetech.
22	Un-authorised access to dock leveller could result in injury to personnel and damage to equipment.	Possible	Catastrophic	• Include this hazard in your risk assessment.	Owner	No
23	Material handling equipment or personnel on Dock Leveller when truck departs could result in injury to personnel or damage to equipment.	Possible	Major	• Provide clear instructions in the Owner's manual that specifies that Dock Leveller must be clear of personnel & equipment when truck departs.	Designer	Yes
				• Ensure that the Dock Leveller is clear of all equipment & personnel when truck departing. • Include this hazard in your risk assessment.	Owner	No
24	Operator attempting to lift the Dock Leveller with equipment or personnel on the Dock Leveller could result in injury to personnel and damage to equipment.	Possible	Major	• Provide clear instructions in the Owner's manual that specifies that Dock Leveller must be clear of personnel & equipment when operating.	Designer	Yes
				• Ensure that the equipment & personnel are clear of Dock Leveller when unit is operating. • Include this hazard in your risk assessment.	Owner	No
25	When cleaning under Dock Leveller and another person operates the dock there is a risk o f a shear/crush injury to the person under the dock.	Possible	Catastrophic	• Provide clear instructions in the Owner's manual that specifies the requirements for cleaning under the Dock Leveller.	Designer	Yes
				• Ensure that procedures for the safe & effective cleaning under Dock Leveller are developed & implemented. • Include this hazard in your risk assessment.	Owner	No
26	Loading equipment driven onto Dock Leveller when Dock leveller is in lowered position and not resting on a truck may cause loading	Possible	Catastrophic	• Provide clear instructions in the Owner's manual that specifies not to drive loading equipment onto Dock Leveller when unit is in lowered position and not resting on the truck bed.	Designer	Yes

	equipment to go out of control resulting in injury to personnel and damage to equipment.			• Ensure that loading equipment is not driven onto Dock Leveller when Dock leveller is in lowered position and not resting on a truck. • Include this hazard in your risk assessment.	Owner	No
27	If auto return to cross traffic installed, when truck departs dock automatically returns to cross traffic position which could result in injury to personnel standing close to the Dock Leveller	Possible	Catastrophic	• Provide clear instructions in the Owner's manual that specifies operation of automatic return to cross traffic.	Designer	Yes
				• Ensure Dock Leveller is clear of personnel when operating. • Include this hazard in your risk assessment.	Owner	No
28	Trucks entering or leaving dock area create a traffic hazard that could result in injury to personnel or damage to equipment.	Possible	Catastrophic	• Develop & implement procedures for traffic management in the docking area. • Include this hazard in your risk assessment.	Owner	No
29	If truck not restrained during loading/unloading truck may move resulting in injury to personnel or damage to equipment.	Possible	Catastrophic	• Provide clear instructions in the Owner's manual that specifies that truck should be restrained during loading/unloading operations.	Designer	Yes
				• The Owner must use suitable restraining equipment and/or procedures to ensure that the truck does not move. • Include this hazard in your risk assessment.	Owner	No
30	Use of a damaged or broken Dock Leveller could result in injury to personnel and further damage to equipment.	Possible	Catastrophic	• Provide clear instructions in the Owner's manual that specifies not to use a damaged or broken Dock Leveller.	Designer	Yes
				• Ensure regular service to be carried out. • If damaged, use effective Lock out/Tag out system and arrange for repair. • Include this hazard in your risk assessment.	Owner	No
31	If load lip does not rest securely on truck bed, the Dock Leveller may fall under load resulting in injury to personnel & damage to equipment.	Possible	Catastrophic	• Provide adequate instruction in Owner's manual that specifies load lip should rest securely on truck bed.	Designer	Yes
				• Ensure that the load lip is resting securely on the truck bed before placing load on the unit. • Include this hazard in your risk assessment.	Owner	No
32	If truck bed is too high or too low, unsafe loading conditions may occur resulting in injury to personnel or damage to equipment.	Possible	Catastrophic	• Provide adequate instruction in Owner's manual	Designer	Yes
				• Ensure that Dock Leveller is not operated if truck bed is too high or too low. • Include this hazard in your risk assessment.	Owner	No

33	Excessive incline on Dock Leveller may result in unsafe forklift operating conditions	Possible	Major	• Provide adequate instruction in Owner's manual	Designer	Yes
				• Include this hazard in your risk assessment.	Owner	No
34	If loading equipment is operated too close to the edge of the Dock Leveller the loading equipment may become unstable and go out of control.	Possible	Catastrophic	• Include this hazard in your risk assessment.	Owner	No
35	Operation of Dock Leveller may not be controlled if operator under the influence of drugs or alcohol and could result in injury to personnel and damage to equipment.	Possible	Catastrophic	• Include this hazard in your risk assessment.	Owner	No
36	High pressure water sprayed on the electrical boxes or control panel may result in electrical faults and potential electrical shock to personnel.	Unlikely	Catastrophic	• Provide adequate instructions in Owner's manual.	Designer	Yes
				• Do not spray high pressure water on electrical boxes or control panel. • Include this hazard in your risk assessment.	Owner	No
37	When reversing up to Dock Leveller truck may hit Dock Leveller resulting in damage to equipment.	Possible	Minor	• Provide adequate instructions in Owner's manual.	Designer	Yes
				• Ensure truck does not make contact with Dock leveller. • Include this hazard in your risk assessment.	Owner	No
38	Forklift speeds exceeding 5 kph result in high impact to the Dock Leveller resulting in damage to equipment.	Possible	Minor	• Provide adequate instructions in Owner's manual.	Designer	Yes
				• Ensure forklift speed does not exceed 5 kph. • Include this hazard in your risk assessment.	Owner	No
39	If manual maintenance prop fitted, raising the maintenance prop whilst under the Dock Leveller may result in injury to personnel.	Possible	Catastrophic	• Provide clear instructions in owner's manual for engaging maintenance prop	Designer	Yes
				• Ensure safety instructions for engaging maintenance prop are adhered to. • Include this hazard in your risk assessment.	Owner	No

40	If hydraulic boom gate is fitted, there is a risk of injury to personnel standing under or close to the boom gate or boom.	Possible	Catastrophic	• Design to incorporate - a pressure limiting valve so that boom gate will stop if light contact made. - flow control valves to limit velocity of boom gate. - solenoid check valves fitted to cylinder to prevent sudden collapse of gate in case of hydraulic failure. • Design to incorporate 85 mm gap between boom gates to eliminate shear/crush point and a 50 mm gap between boom gate & boom to eliminate shear/crush point. • Provide warning sign near boom gate. • Provide clear instructions in Owner's manual to stand clear when boom gate operating	Designer	Yes
				• Include this hazard in your risk assessment.	Owner	No
41	If hydraulic boom gate is fitted, there is a risk of injury to personnel standing under or close to the boom gate or boom.	Possible	Catastrophic	• Design to incorporate - a pressure limiting valve so that boom gate will stop if light contact made. - flow control valves to limit velocity of boom gate. - solenoid check valves fitted to cylinder to prevent sudden collapse of gate in case of hydraulic failure. • Design to incorporate 85 mm gap between boom gates to eliminate shear/crush point and a 50 mm gap between boom gate & boom to eliminate shear/crush point. • Provide warning sign near boom gate. • Provide clear instructions in Owner's manual to stand clear when boom gate operating	Designer	Yes
				• Include this hazard in your risk assessment.	Owner	No
42	If manual boom gate is fitted, when boom gate is being operated, there is a risk of injury to personnel standing under or near to the boom gate or boom.	Possible	Major	• Design to incorporate system to control lowering speed using gas air struts	Designer	Yes
				• Include this hazard in your risk assessment.	Owner	No

43	If manual boom gate is fitted, there is a possibility of damage to gas air strut. This might cause the booms to fall.	Possible	Major	• Gas struts enclosed within boom gate structure thus removing risk of damage	Designer	Yes
				• Gas struts to have regular maintenance inspection. • Include this hazard in your risk assessment.	Owner	No
44	If Dock Leveller is fitted with an intrinsically safe maintenance prop, incorrect/uncontrolled operation of maintenance prop release arm could result in a crush injury.	Possible	Catastrophic	• Provide clear instructions in owner's manual for operation of release arm.	Designer	Yes
				• Ensure safety instructions for operation of release arm are adhered to. • Include this hazard in your risk assessment.	Owner	No
45	If Dock leveller is fitted with an intrinsically safe maintenance prop, the threaded bolts holding the maintenance prop could break causing the dock leveller to lower. This could result in a shear/crush injury.	Possible	Catastrophic	• Fit a Velocity Control device to prevent sudden collapse of unit, and ensure specifications of manufacture comply with relevant standards. • Provide clear instructions in the Owner's manual that specify the requirements for access under the unit.	Designer	Yes
				• Develop & implement a safe work procedure for cleaning under the dock. • Include this hazard in your risk assessment.	Owner	No
46	If Dock Leveller is fitted with an intrinsically safe maintenance prop, if the arm of the maintenance prop that supports the lip breaks, the lip will fall which could possibly result in injury to personnel.	Possible	Catastrophic	• Provide clear instructions in the Owner's manual that specify the requirements for access under the unit.	Designer	Yes
				• Develop & implement a safe work procedure for cleaning under the dock. • Include this hazard in your risk assessment.	Owner	No
47	If the Dock Leveller is fitted with an intrinsically safe maintenance prop, if the springs on the maintenance prop release arm break, part of the spring could be forcibly propelled and could possible cause injury to personnel.	Possible	Major	• Include this hazard in your risk assessment.	Owner	No

7.2 Risk Matrix

Risk = Probability (likelihood) x consequence
Risk score is calculated by multiplying the points assigned to the probability (likelihood) by the points assigned to the consequence.
Risk score is indicative of, if action is required to reduce the risk, and when action is to be done.

				Consequence				
				Catastrophic	Major	Moderate	Minor	Insignificant
				Death	Extensive injuries- Amputation	Medical treatment - Doctor	Minor-First Aid	No Injuries
			Points	40	15	7	3	1
Likelihood	**Almost Certain**	Expected to occur in most circumstances	100	4000	1500	700	300	200
	Likely	Probably occur in many circumstances	36	1440	540	252	108	36
	Possible	Might occur at some time	9	360	135	63	27	9
	Unlikely	Could occur at some time	2	80	30	14	6	2
	Rare	May occur in exceptional circumstances	1	40	15	7	3	1

7.3 Risk Score Rating:

Risk Score (=Probability points consequence points)	Risk Rating	Action
More than 320	Very High	Immediate
160-320	High	Prompt
70-160	Significant	Correction required
40-70	Moderate	Attention is needed
Less than 40	Low	Perhaps acceptable

8.0 Drawings

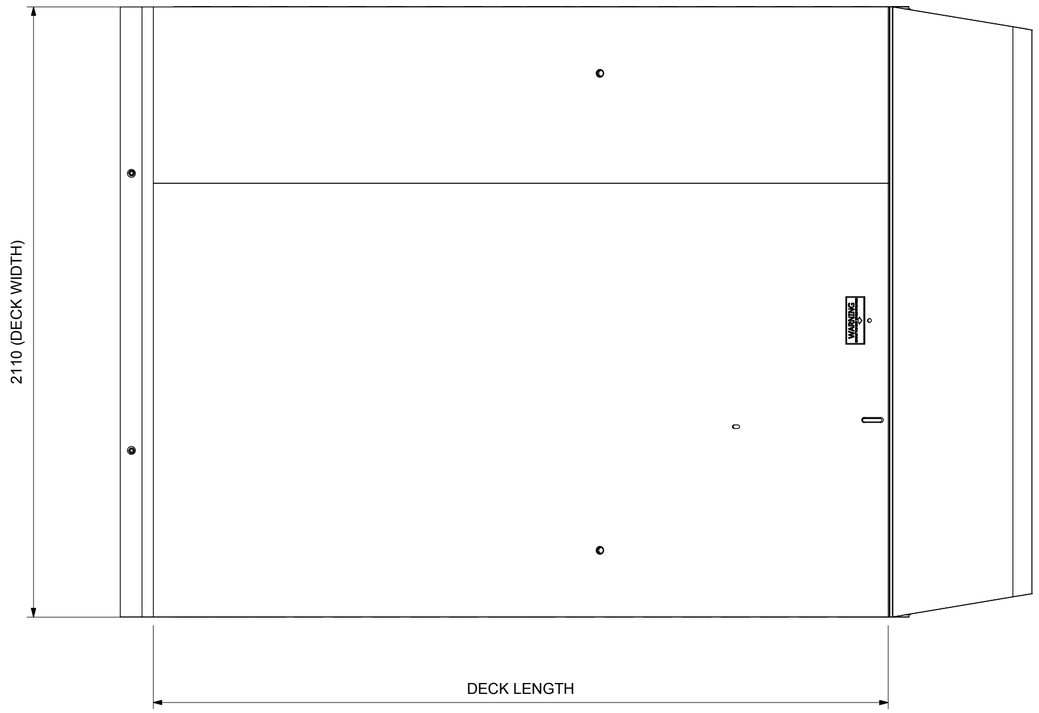
Attachment 1	As Built General Arrangement Drawings
Attachment 2	Electrical Schematics

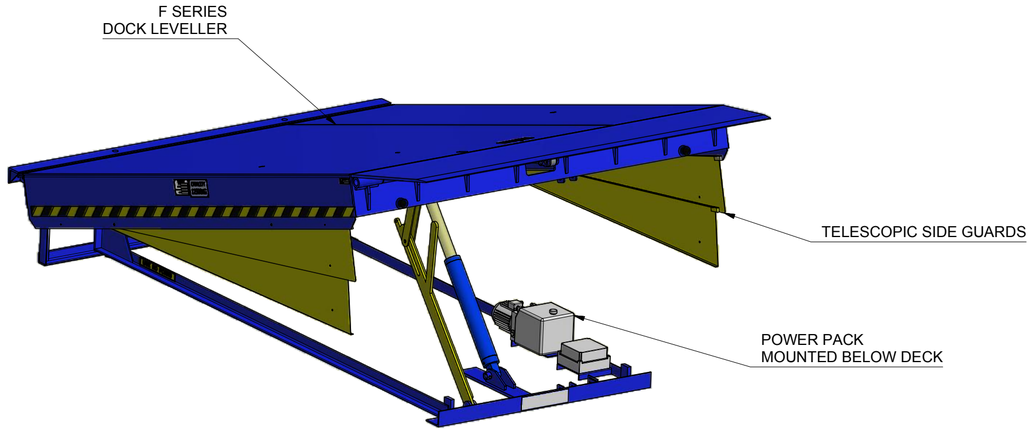
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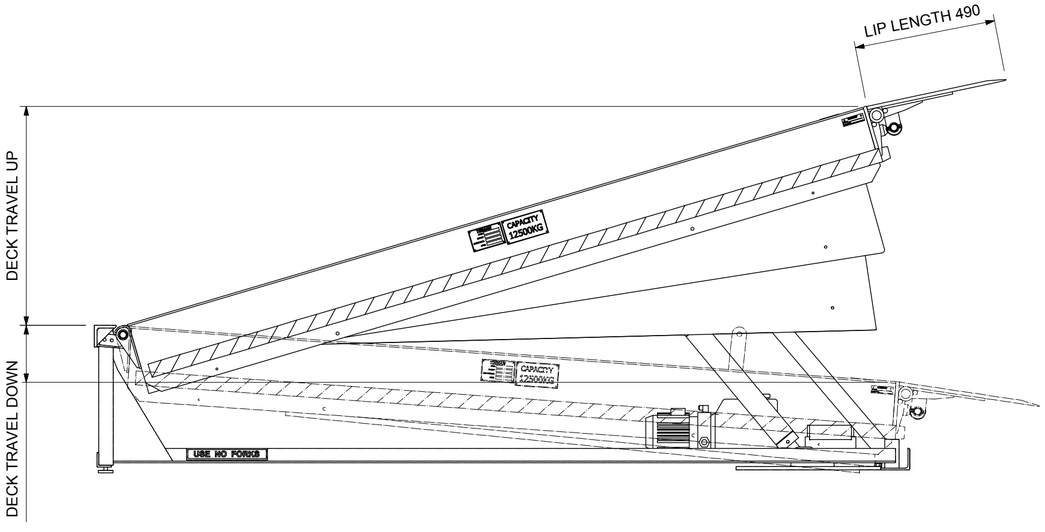
AS BUILT

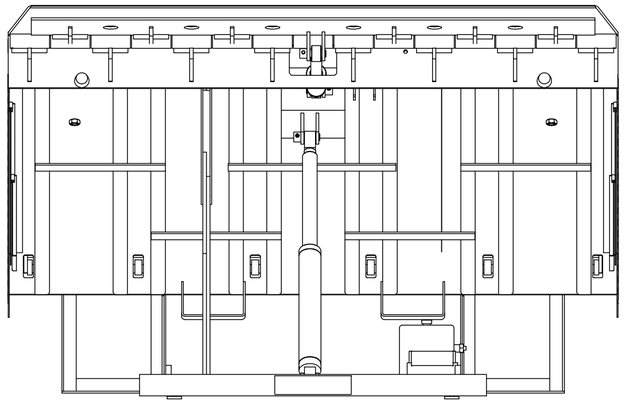
MODEL	DECK LENGTH (mm)	TRAVEL UP (mm)	TRAVEL DOWN (mm)
703	600	350	75
708	2210	750	180
710	2820	750	180
712	3440	1000	180
714	4040	1075	180





PERSPECTIVE VIEW
(1 : 20)





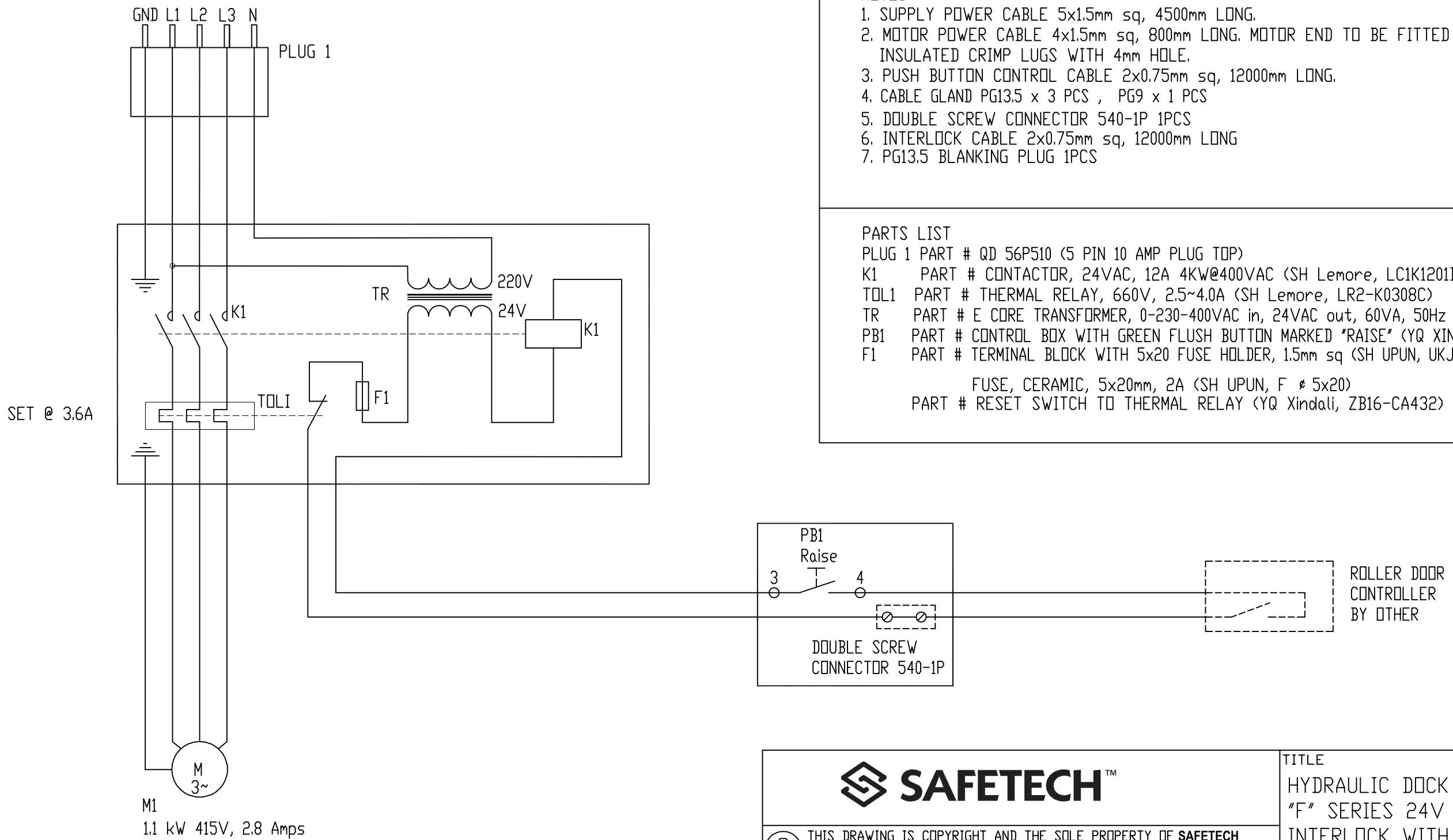
S:\Products\Dock Products\Dock Levellers\Engineering\Dwgs & Tech Info\Dock Levellers FAME CAD\710\DOCK-LEVELLER.700-GA.idw - Sheet # 1

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QUANTITY:					FINISH:	
ALL DIMENSIONS IN mm UNLESS STATED OTHERWISE	CODE:	DRAWN: rossf	DRG. NO.:	JOB NO.:	TITLE: **700 F SERIES DOCK LEVELLER** GA - 703/708/710/712/714	
FILENAME: DOCK-LEVELLER.700-GA.idw	SCALE: NTS	DATE: 16/08/2016	SIZE: A2	DXF FILE:	SERIAL NO.:	

REVISIONS			
REV	DETAILS OF CHANGE	DRN	DATE
A	NOTE 5 & CONNECTOR ADDED	LT	07-08-2014
B	REVISED DOOR CONTROLLER AND CHANGED SCREW CONNECTOR. REVISED NOTES & PART LIST	LT	24-09-2014



NOTES:
1. SUPPLY POWER CABLE 5x1.5mm sq, 4500mm LONG.
2. MOTOR POWER CABLE 4x1.5mm sq, 800mm LONG. MOTOR END TO BE FITTED INSULATED CRIMP LUGS WITH 4mm HOLE.
3. PUSH BUTTON CONTROL CABLE 2x0.75mm sq, 12000mm LONG.
4. CABLE GLAND PG13.5 x 3 PCS , PG9 x 1 PCS
5. DOUBLE SCREW CONNECTOR 540-1P 1PCS
6. INTERLOCK CABLE 2x0.75mm sq, 12000mm LONG
7. PG13.5 BLANKING PLUG 1PCS

PARTS LIST
PLUG 1 PART # QD 56P510 (5 PIN 10 AMP PLUG TOP)
K1 PART # CONTACTOR, 24VAC, 12A 4KW@400VAC (SH Lemore, LC1K1201B7)
TOL1 PART # THERMAL RELAY, 660V, 2.5~4.0A (SH Lemore, LR2-K0308C)
TR PART # E CORE TRANSFORMER, 0-230-400VAC in, 24VAC out, 60VA, 50Hz (ZJ ZHONGCE ETE09205RD)
PB1 PART # CONTROL BOX WITH GREEN FLUSH BUTTON MARKED "RAISE" (YQ XINDALI, LAY5-B103P)
F1 PART # TERMINAL BLOCK WITH 5x20 FUSE HOLDER, 1.5mm sq (SH UPUN, UKJ-2.5RD)
FUSE, CERAMIC, 5x20mm, 2A (SH UPUN, F φ 5x20)
PART # RESET SWITCH TO THERMAL RELAY (YQ Xindali, ZB16-CA432)

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DRAWN	JD	NEXT ASSY	-
DATE	31-01-2012	GEN ARRG'T	-
APPR'D	-	TOLERANCES	-
SCALE	-		

TITLE
HYDRAULIC DOCK LEVELLER
"F" SERIES 24V CONTROL
INTERLOCK WITH ROLLER DOOR

CLIENT
-
-

DRAWING NO 6A248-07

REV B